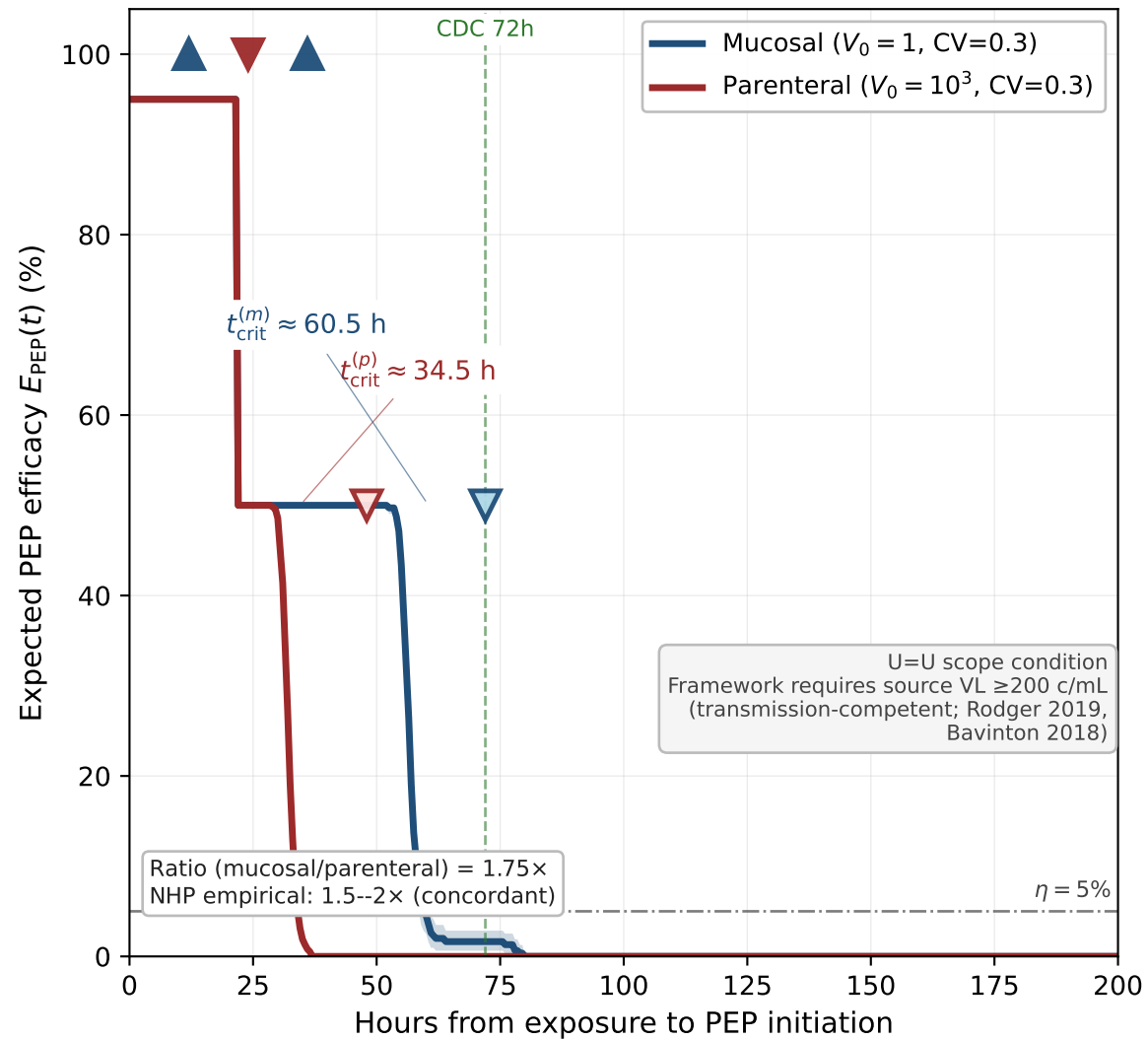


A. Route comparison: emergent t_{crit} from multiscale dynamics

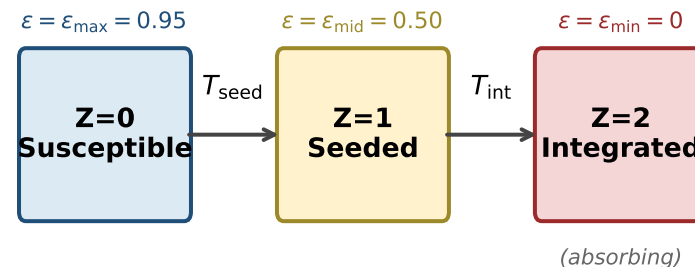


NHP empirical ($\blacktriangle$ concordant, $\blacktriangledown$ not concordant)

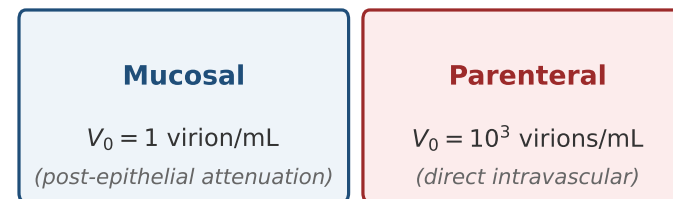
$\blacktriangle$ Mucosal, model concordant (NHP=100%) $\blacktriangledown$ Parenteral, not concordant (NHP=100%)

$\triangleleft$ Mucosal, not concordant (NHP=50%) $\triangleright$ Parenteral, not concordant (NHP=50%)

B. Three-state within-host model & route-dependent inoculum



Route-dependent inoculum



By inoculum-monotone hitting times (Lemma S5.2): $T_{\text{int}}^{(p)} \leq T_{\text{int}}^{(m)}$

$\Rightarrow t_{\text{crit}}^{(p)} \leq t_{\text{crit}}^{(m)}$ (Corollary S5.1)